

Original Article

Microbiome dynamics and genomic determinants of bovine mastitis

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ABSTRACT

The milk of lactating cows presents a complex ecosystem of interconnected microbial communities which can influence the pathophysiology of mastitis. We hypothesized possible dynamic shifts of microbiome composition and genomic features with different pathological conditions of mastitis (Clinical Mastitis; CM, Recurrent CM; RCM, Subclinical Mastitis; SCM). To evaluate this hypothesis, we employed whole metagenome sequencing (WMS) in 20 milk samples (CM, 5; RCM, 6; SCM, 4; H, 5) to unravel the microbiome dynamics, interrelation, and relevant metabolic functions. The WMS data mapped to 442 bacterial, 58 archaeal and 48 viral genomes with distinct variation in microbiome composition (CM > H > RCM > SCM). Furthermore, we identified a number of microbial genomic features, including 333, 304, 183 and 50 virulence factors-associated genes (VFGs) and 48, 31, 11 and 6 antibiotic resistance genes (ARGs) in CM, RCM, SCM, and H-microbiomes, respectively. We also detected different metabolic pathway and functional genes associated with mastitis pathogenesis. Therefore, profiling microbiome dynamics in different conditions of mastitis and associated microbial genomic features contributes to developing microbiome-based diagnostics and therapeutics for bovine mastitis.

1. Introduction

Mastitis is the inflammation of the mammary gland and/or quarters [1] and represents the foremost disease-driven challenge in production faced by the global dairy industry [2]. The milk microbiome composition is an important determinant of mammalian health [3,4], plays an important role in udder health by interacting with the immune and metabolic functions of the cow [5,6], the spread of virulence factors [7], and the spread of antimicrobial resistance genes (ARGs) [8,9]. Previously, we reported that the microbiome signature in bovine clinical mastitis (CM) has distinct shifts and differences in abundance between the microbiome of CM and H milk [6]. Furthermore, bovine CM milk had 68.04% inclusion of previously unreported or opportunistic pathogens [6]. In addition, a plethora of archaeal and viral species [6] were also components of the microbiome, suggesting relevant physiological and pathological implications for their host. Although the

bovine mastitis microbiome is comprised of bacteria, archaea, viruses, and other microorganisms, until now, most researchers have mainly focused on the bacterial component of the milk microbiome with limited targeted amplicon (16S) sequencing [10–13].

We also reported that the resistomes (the total collection of antibiotic resistance genes) act as a potential key factor in disease complication and recurrence, could be concurrent to microbiome signature [14]. However, no association was found with cattle breed, despite significant differences in microbiome diversity among different breeds [14]. The taxonomic diversity and composition of microbiomes in bovine mastitis may also vary according to different forms of mastitis (clinical, subclinical and recurrent) as also reported earlier in several studies [6,10,13,15,16]. Following pathogen invasion and subsequent establishment in the mammary gland, either CM or subclinical mastitis (SCM) may present itself. Bovine CM is one of the most frequently observed diseases affecting the global dairy industry with visibly

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abnormal milk, swollen, hot, painful, and reddened udders [17,18]. The prevalence rate of CM ranges from 13.0 to 50.0% in different countries and cattle breeds [17, 18, [19]. Moreover, diverse groups of microbial communities colonizing the mammary gland or quarters have evolved novel mechanisms to facilitate their proliferation during the pathogenesis of CM. This diversity comprises both contagious udder pathogens including *Staphylococcus aureus*, *Streptococcus agalactiae*, *Streptococcus dysgalactiae*, *Mycoplasma* spp., and *Corynebacterium bovis* [6,7,10], and environmental pathogens such as *Escherichia coli*, *Klebsiella pneumoniae*, *Klebsiella oxytoca*, *Enterobacter aerogenes*, *Streptococcus dysgalactiae*, and *Streptococcus uberis* [6,7,10,11]. One of the very frustrating aspects of bovine CM is its recurrent nature, reinfection of a quarter or udder after bacteriological cure [20]. Recurrent CM (RCM) is caused by persistent intra-mammary infections (IMI) [18], and specific differences in the milk microbiome may contribute to recurrence susceptibility [20]. IMI may persist beyond the resolution of the clinical symptoms of CM, and subsequent RCM flare-ups may be observed [18]. RCM associated with persistence of pathogens in the udder *E. coli*, *S. aureus*, *S. uberis*, and *S. dysgalactiae* was limitedly reported [18]. Almost every dairy herd has cows with SCM [15], and a variety of pathogens can establish chronic infections which may only occasionally manifest the clinical signs of mastitis [21]. The incidence of SCM remains higher than CM, and can range from 25.0 to 78.5% in different breeds and countries [15,19]. However, unlike CM, the SCM in dairy herds always remained unnoticed due to the absence of clinical symptoms [19]. Predominantly identified SCM pathogens include *E. coli*, *Pseudomonas aeruginosa*, *P. mendocina*, *Shigella flexneri*, *Bacillus cereus*, *K. pneumoniae*, *S. chromogenes*, and *S. epidermidis* [10,11,15].

Though not studied well, we hypothesize that during the progression of mastitis, dysbiosis in the milk microbiome can occur, with an increase in opportunistic pathogenic bacteria, and a reduction in healthy milk commensal bacteria [6,22,23]. Several earlier studies reported that *S. aureus*, coagulase-negative *Staphylococci* (CNS), *S. agalactiae*, *S. dysgalactiae*, *S. uberis*, *Enterococcus* spp., *Arcanobacterium pyogenes*, *Pseudomonas* spp., *Micromonospora* spp., and *Klebsiella pneumoniae* are environmental pathogens that play a role as the potential opportunists in the pathophysiology of bovine mastitis [6,24]. Conversely, *Acinetobacter*, *Aerococcus*, *Corynebacterium*, *Enterobacter*, *Facklamia*, *Lactobacillus*, *Lactococcus*, *Micrococcus*, *Propionibacterium*, *Staphylococcus*, *Streptococcus*, and non-aureus *Staphylococci* (NAS) are the so far reported commensal bacteria of bovine milk [7]. Researchers have now started to explore the complexities of the microbial ecology within this niche through disease progression [6,7,14,22]. While understanding the role of all interacting microbial players contributing to disease progression is becoming increasingly crucial in the face of large-scale epidemics [25,26], such a study has yet to be completed for dairy animal mastitis. The microbial community present within the mammary gland and/or quarters is being increasingly recognized as a key regulator of metabolic and immune homeostasis [5,6], and a mediator of antimicrobial resistance [6]. The resistomes or ARGs, which exist in both pathogenic [9,14] and nonpathogenic commensal bacteria [9] are frequently carried on mobile genetic elements. Although, studying the intrinsic resistance of the bovine mastitis microbiome would provide knowledge on the mechanism, and the possible treatment to tackle the disease, extensive study on the resistome and/or ARGs has not been carried out yet. Data are limited to individual isolate level either through in vitro assays or genomic characterization [9,14,16,18]. Similarly, the virulome, the set of genes encoding virulence, can also be carried on the mobilome [9], and thus, many virulence factor associated genes (VFGs) easily spread in bacterial populations by horizontal gene transfer, converting mutualistic or commensal bacteria into potential opportunistic pathogens [7,23,27]. However until now, the role of different VFGs such as genes coding for capsular activity (ABZJ), nutrient (iron) uptake (*pvdL*), twitching motility and/or chemotaxis (*chpA* and *motC*), bacterial adhesion, invasion, or intracellular survival

(*motB/ompA*, *pilJ* and *pvdA*), two-component system (*gacS*), intracellular multiplication factor (*icmF1*), biofilm formation and quorum sensing (*bfnR* and *pvdQ*), heat shock or stress (*mucD*), bacterial communication (*hdtS*), and, and bacterial pathogenesis (*crc*) have been studied for individual mastitis pathogens [1,5,16,24], but not studied at microbiome level. Though the exact functions of archaea and viruses in the progression of mastitis are not fully explored like bacterial pathogens [6,14,17], the dynamic shifts in their composition and relative abundances at the onset of CM, persistent or chronic recurrent CM (RCM), invisible udder infections (SCM), and the healthy conditions of the mammary glands and/or quarters could result from a host genetic predisposition to colonization, immune responses to milk bacteria involving increased levels of polymorphonuclear infiltrate, differences mediated by the microbial composition itself, or a combination of these effectors [6,11,27]. However, it is not fully understood what exactly happens to the microbiome when the disease state changes; which microbial species are responsible, and what properties enable them to persist through different disease states. Consequently, treatment of disease could be exceedingly difficult rendering therapeutics ineffective to alleviate disease. Therefore, we adopted a comprehensive study encompassing all the mastitis categories and investigated their dynamics and associated microbial genomic pathogenic determinants (genomic features).

We employed the high-throughput WMS technique (on an average 20.83 million reads per sample, Data S2 & S2) to gain insight into the phylogenetic composition and diversity of these microbiomes [6,28], and to profile their functional attributes [6,29] in 15 mastitis (CM, RCM, SCM) and 5 healthy (H) milk samples. We investigated whether consistent and specific disease-associated changes in the microbial communities, and their genomic features could be identified across the sample categories of bovine mastitis (CM, RCM and SCM) and healthy milk. Culture-independent WMS investigation of the bovine milk microbiome suggested the presence of diverse microbes in milk from healthy and infected mammary glands far greater than previously described [6,11,30].

We propose, there are possible dynamic shifts of microbiome compositions in different types mastitis under different pathophysiological states determined by its favoring genomic potentials including VFGs, ARGs, and metabolic functions. Moreover, the overexpression of genes related to ARGs, VFGs, and several metabolic pathways coming from the metabolic activities of the microbiome is a crucial factor for the development and progression of bovine mastitis. Therefore, our study aims to describe, for the first time, a comprehensive scenario of the alterations in microbiome composition and test for associations with microbial genomic features, including, VFGs, ARGs, and metabolic functions across different pathological categories of bovine mastitis (CM, RCM, SCM), and healthy milk through the WMS approach.

2. Results

2.1. Mastitis-associated changes in the milk microbial diversity and community structure

To evaluate the effect of bovine mastitis categories (CM, RCM, SCM) on the milk microbial alpha diversity (i.e., within-sample diversity), we computed the number of species and/or strains observed in each sample (i.e., richness), and Chao1, ACE, Shannon, Simpson and Fisher indices (i.e., a diversity index accounting both evenness and richness) at the strain level through PathoScope 2.0 (PS) [31] analysis. We found that both the number of observed species/strains, Chao1, ACE, Shannon, Simpson and Fisher indices estimated diversity significantly varied across the four sample groups (CM, RCM, SCM and H) ($p = 0.003$, Kruskal-Wallis test), and were higher in CM and RCM groups compared to the groups of SCM and H-milk samples (Fig. 1a, b). The rarefaction curves showed that the depth of coverage was sufficient to capture the entire microbial diversity within each sample (Fig. S1, Data S2).

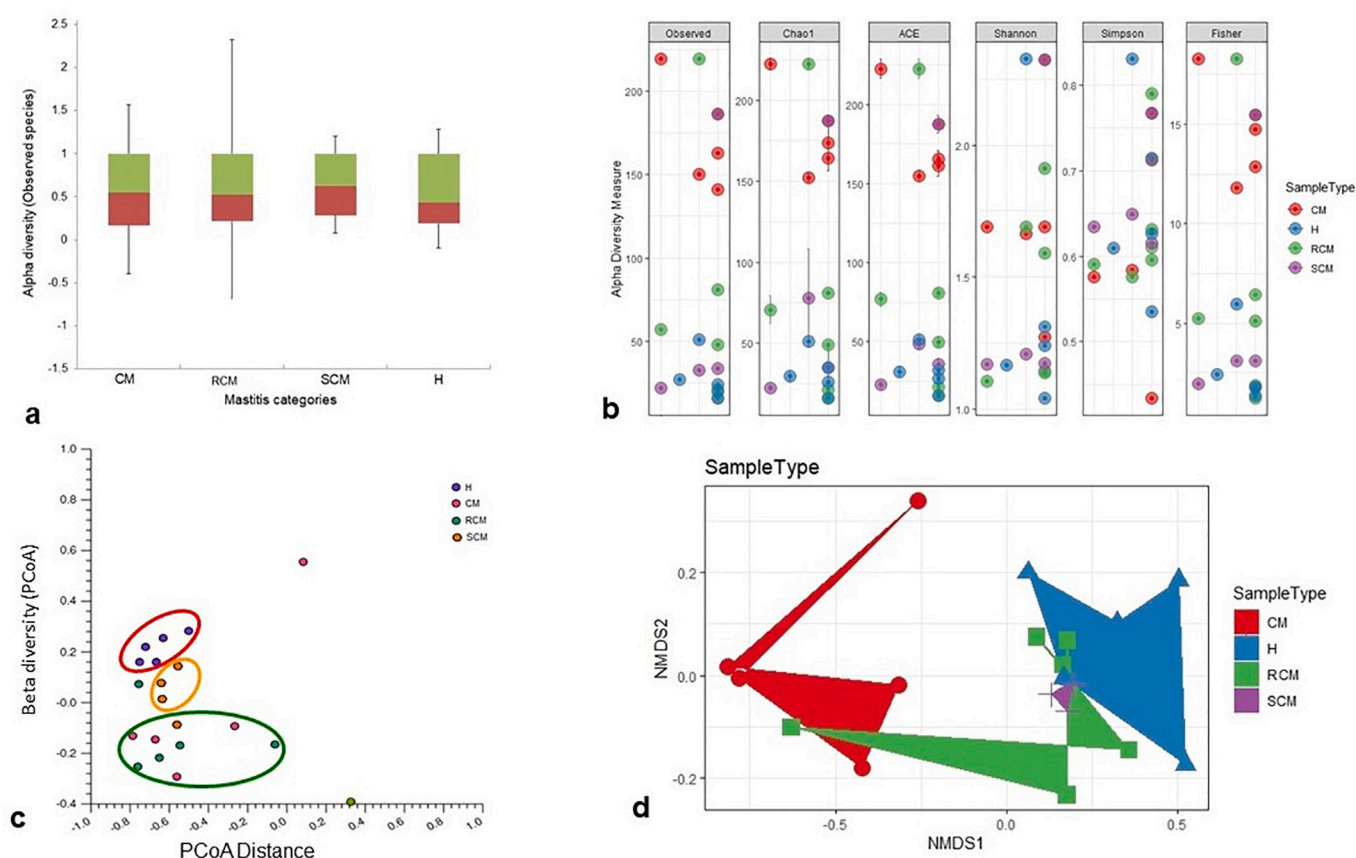


Fig. 1. Differences in microbiome community between bovine mastitis and healthy milk. (a) Box plots showing significant ($P_{\text{Observed}} = 0.032$) differences in observed species and/or strains richness in clinical mastitis (CM), recurrent clinical mastitis (RCM), subclinical mastitis (SCM) and healthy (H) milk samples. (b) Alpha diversity measured using the observed species, Chao 1, ACE, Shannon, Simpson and Fisher diversity indices through PathoScope (PS) analysis revealed distinct microbiome diversity across the samples of four metagenomes ($p = 0.003$, Kruskal-Wallis test). (c) Principal coordinates analysis (PCoA) measured on the Bray-Curtis distance method using MG-RAST (MR) tool (genus-level) colored samples by populations. Each dot represents an individual, and colors indicate the populations in four metagenomes. (d) Non-metric multidimensional scaling (NMDS) ordination plots showing the clear separation between CM, RCM, SCM and H milk samples as measured by weighted-UniFrac distance on PS data at strain level. In both cases, the CM microbiomes clustered more closely to RCM microbiomes (green circle in c, and red and green colored zones in d) followed by SCM microbiomes (red circle in c and violet zone in d), and however, more distantly clustered with H milk microbes (pink circle in c, and blue zone in d). Statistical analysis using Kruskal-Wallis tests showed significant microbial diversity variations across the four metagenomes ($p = 0.001$, Kruskal-Wallis test). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Principal coordinate analysis (PCoA), as measured on the Bray-Curtis distance method (Fig. 1c) using MG-RAST (MR) [32], and non-metric multidimensional scaling (NMDS) ordination plots, through weighted-UniFrac distance method (Fig. 1d), showed discrimination across the metagenomes. The CM-microbiomes clustered more closely to the RCM-microbiomes followed by SCM-microbiomes, and more distantly clustered with H-milk microbes (Fig. 1c, d). Therefore, significant variation in microbiome diversity and composition across these four categories ($p = 0.001$, Kruskal-Wallis test, Beta diversity test Fig. 1 c, d) was also evident. Furthermore, by annotating the WMS reads against different reference databases (namely, virulence factor database (VFDB) [33], ResFinder [34], and KEGG and SEED modules of MR [32,35], we demonstrated that there were distinct differences in mastitis-specific factors such as VFGs, ARGs, and metabolic functions (Data S3) in four metagenomes, which could strongly modulate microbiomes dysbiosis, and the pathophysiology of bovine mastitis.

The composition of bovine milk microbiomes at the domain level was numerically dominated by bacteria, with a relative abundance of 99.58%, followed by archaea (0.24%), and viruses (0.18%) (Data S2). In this study, the relative abundances of the microbiomes (bacteria, archaea and viruses) always remained higher in mastitis (CM, RCM and SCM)-metagenomes compared to the H-milk metagenome. However, the relative abundances of microbes fluctuated more within mastitis

groups. The unique and shared distribution of microbiota found in the mastitis and healthy-milk samples are represented by comprehensive Venn diagrams (Fig. 2). We detected 18 bacterial phyla, of which 16, 11, 4 and 13 were found in CM, RCM, SCM and H milk metagenomes, respectively (Data S2). Moreover, four bacterial phyla (*Proteobacteria*, *Firmicutes*, *Actinobacteria* and *Bacteroidetes*) were shared across the four groups (Fig. S2). The CM milk metagenome had an exclusively unique association with five bacterial phyla (*Deinococcus*, *Acidobacteria*, *Chrysiogenetes*, *Deferribacteres* and *Chloroflexi*), and two typical bacterial phyla (*Thermotogae* and *Verrucomicrobia*) were uniquely present in H-milk samples (Data S2). Moreover, 78 orders of bacteria were identified in four metagenomes, of which 13.89%, 4.17% and 6.56% orders were exclusively unique in CM, RCM, and H-metagenomes, respectively (Fig. S3, Data S2). The MR pipeline simultaneously detected bacteria, archaea, and viruses in these metagenomes. A total of 373 bacterial genera, including 314, 187, 66 and 272 in CM, RCM, SCM and H milk samples, respectively, were detected (Fig. 2a, Data S2), and 18.69%, 3.63%, 4.54%, and 9.75% detected genera had sole associations with CM, RCM, SCM and H conditions of the mammary glands. However, 12.06% bacterial genera (Fig. 2a) were common in all of four sample groups (Data S2). We detected 442 bacterial strains through PS analysis, of which 385, 65, 80 and 144 strains were found in CM, RCM, SCM and H milk, respectively (Fig. 2b). Of the identified strains,

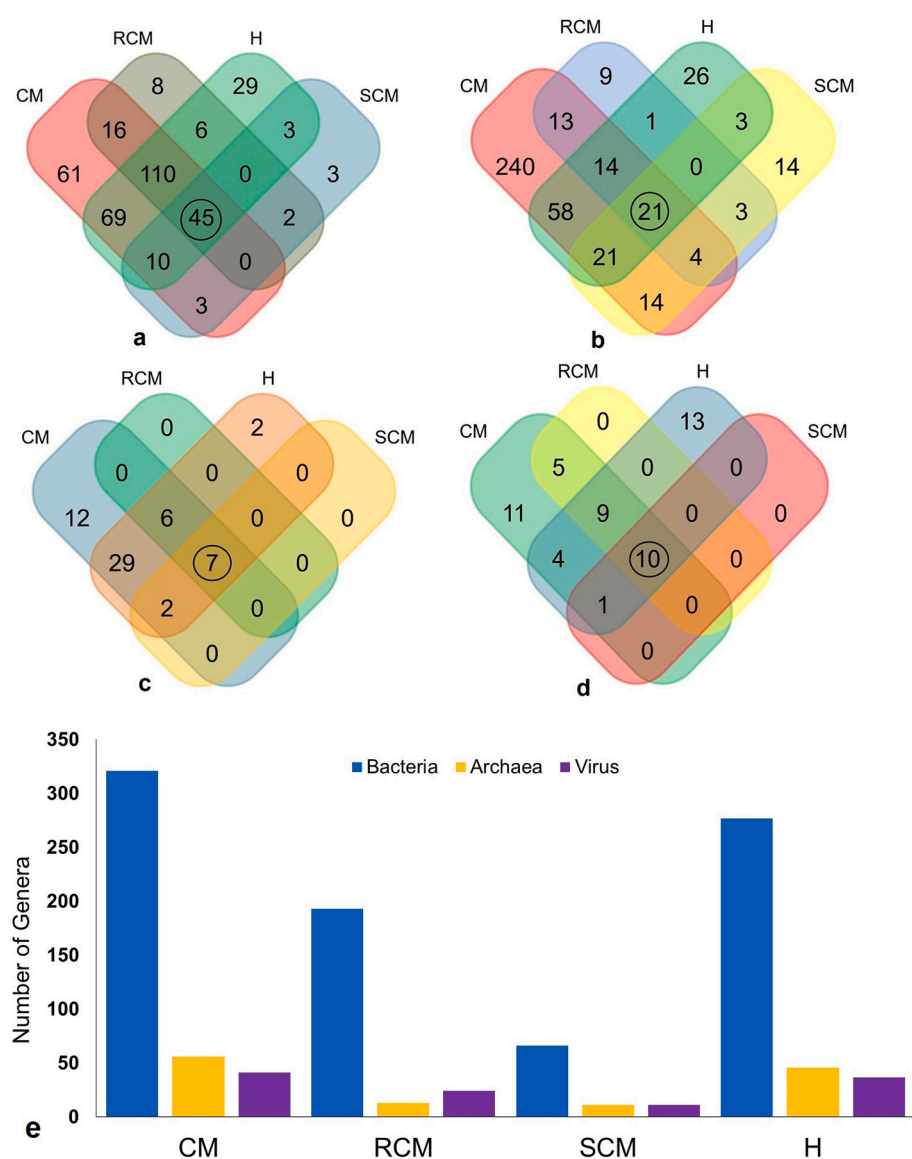


Fig. 2. Taxonomic composition of bovine mastitis and healthy milk microbiome. Venn diagrams representing the core unique and shared microbiomes in CM, RCM, SCM and H milk samples. (a) Venn diagram showing unique and shared bacterial genera by MR analysis, (b) Venn diagram comparison of bacteria at strain level as measured through PS analysis, (c & d) Venn diagrams representing unique and shared archaeal and viral genera, respectively found in mastitis (CM, RCM, SCM) and H milk samples as analyzed with MR pipeline. Microbiome sharing between the conditions are indicated by black circles. (e) Dynamic changes in the composition of bacteria, archaea and viruses in four metagenomes at genus level. Each bar plot represents total genera of bacteria, archaea and viruses detected in the respective metagenome.

15.44%, 12.30%, 3.75% and 20.14% strains in CM, RCM, SCM, and H milk, respectively had exclusive associations, and only 4.75% were found to be shared across these metagenome (mastitis and healthy) samples (Fig. 2b). The CM milk microbiome however had 13.1%, 14.8% and 27.6% shared bacterial strains with RCM, SCM and H microbiomes, respectively (Fig. 2b, Data S2).

The MR pipeline detected 58 archaeal, and 48 viral genera in four metagenomes, and of them, 12.06% archaeal (Fig. 2c), and 20.83% viral (Fig. 2d) were common in four metagenomes. By comparing these genera across the sample categories, we identified 56, 13, 9 and 46 archaeal, and 40, 24, 11 and 37 viral genera in CM, RCM, SCM and H-milk metagenome, respectively. We found that 23.2%, 16.1% and 75.9% archaeal (Fig. 2c, Data S2), and 60.0%, 27.5% and 45.3% viral genera in RCM, SCM and H samples, respectively shared with those of CM samples (Fig. 2d, Data S2). The CM milk metagenome had sole association of 12 (21.43%) archaeal, and 11 (27.50%) viral genera. The composition of the viral and archaeal fractions of the microbiomes remained much higher in CM milk followed by H, RCM, and SCM-milk metagenomes (Fig. 2e). Despite having lower taxonomic composition in RCM and SCM-metagenomes, the relative abundances of the associated microbiota varied significantly ($p = 0.027$, Kruskal-Wallis test) across these microbiomes (Data S2).

2.2. Mastitis-associated shifts in bacteriome composition

We found that both the composition and the relative abundances of bacterial taxa at the phylum, order, genus, and strain-level differed between mastitis and healthy controls irrespective of the bioinformatics tools used. The associated bacterial phyla were numerically dominated by *Proteobacteria*, *Firmicutes*, *Actinobacteria* and *Bacteroidetes* (comprising > 99% of relative abundance) in mastitis and healthy-milk microbiomes. Additionally, *Proteobacteria* was found to be the single most dominating phylum in CM, RCM and H milk samples (> 90% relative abundance), while in the SCM-milk, the most abundant bacterial phylum was *Firmicutes* (42.95%) followed by *Bacteroidetes* (33.95%), *Proteobacteria* (22.15%), and *Actinobacteria* (0.95%) (Data S2). *Pseudomonadales* was the predominantly abundant bacterial order in CM (86.25%), RCM (90.93%) and H (98.51%) milk, whereas *Lactobacillales* (41.59%) and *Flavobacteriales* (31.01%) were the most abundant orders in SCM-milk (Fig. S4, Data S2). The relative abundance of *Enterobacteriales* also varied among the sample categories and remained significantly higher in the mastitis milk (CM, 12.81%; RCM, 5.42%; SCM, 12.11%) compared to the H milk (0.23%) samples. Although the SCM milk had the lowest number of bacterial orders, their relative abundances remained higher than those detected in the other

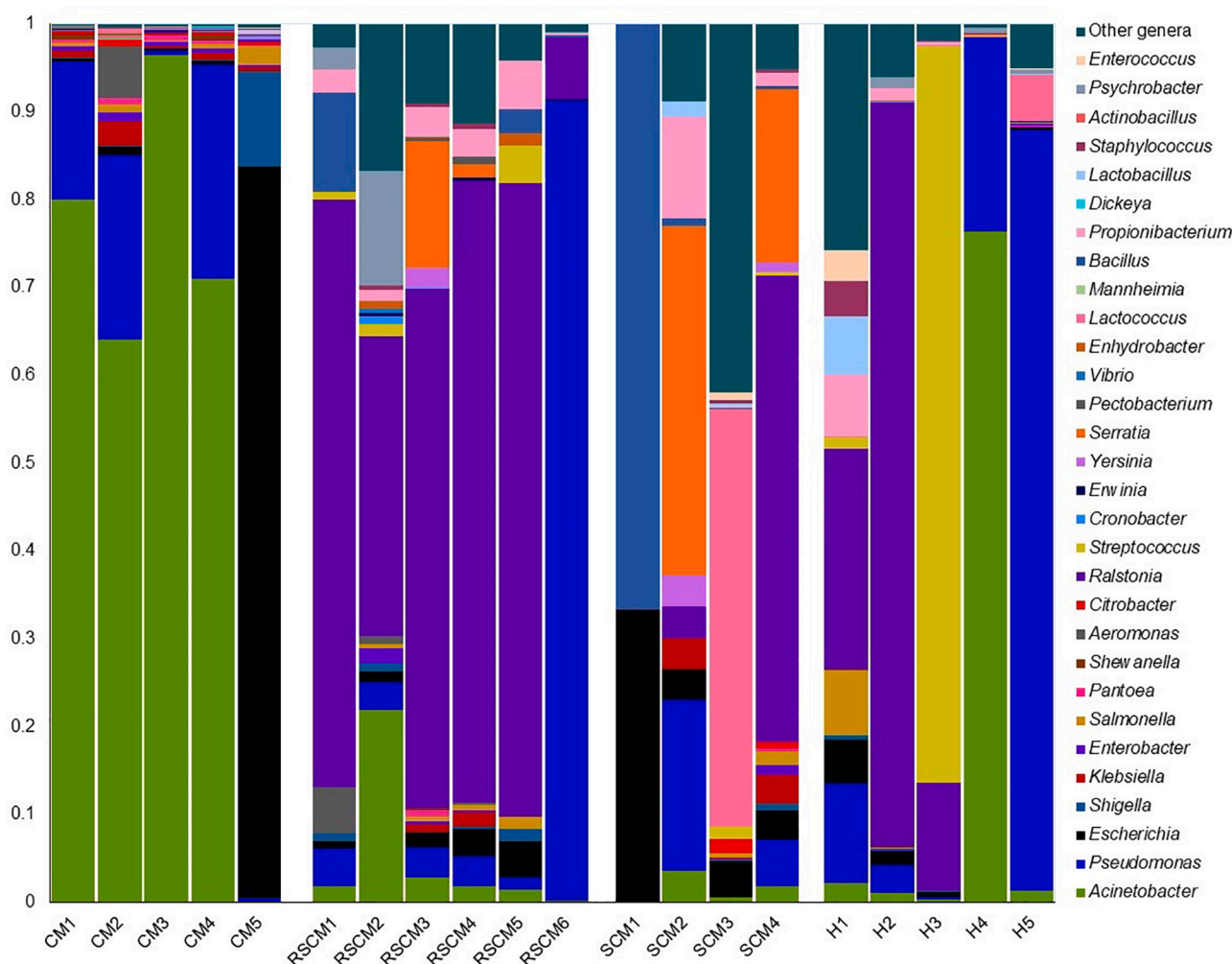


Fig. 3. The genus level taxonomic profile bacteria in bovine mastitis and healthy milk. Stacked bar plots showing the relative abundance and distribution of the 30 most abundant genera, with ranks ordered from bottom to top by their increasing proportion among the CM, RCM, SCM and H milk samples. Only the 29 most abundant genera are shown in the legend, with the remaining genera grouped as 'Other genera'. Each stacked bar plot represents the abundance of bacteria in each sample of the corresponding category. The distribution and relative abundance of the bacterial genera in four metagenomes are also available in Data S1.

samples (CM, RCM, and H) (Fig. S2, Data S2).

We also found significant differences ($p = 0.001$, Kruskal-Wallis test) in the composition and relative abundance of the bacteria at the genus level. In CM and RCM samples, *Acinetobacter* was the most abundant bacterial pathogen with a relative abundance of 68.59% and 72.61%, respectively, but remained much lower (0.81%) in SCM-milk samples. The CM milk microbiome was also predominated by *Pseudomonas* (17.69%), *Escherichia* (8.62%), *Shigella* (1.12%), *Klebsiella* (0.66%), *Salmonella* (0.52%), *Enterobacter* (0.50%), *Shewanella* (0.46%), *Pantoea* (0.46%), and *Citrobacter* (0.32%), and rest of the genera had < 0.1% relative abundance (Fig. 3, Data S2). Simultaneously, *Pseudomonas* (18.21%), *Aeromonas* (2.64%), *Klebsiella* (1.68%), *Enterobacter* (0.79%), *Escherichia* (0.68%), *Pantoea* (0.61%), *Citrobacter* (0.54%), *Salmonella* (0.52%), *Shewanella* (0.35%), *Lactococcus* (0.24%), and *Cronobacter* (0.13%) were the most abundant genera in the RCM (Fig. 3, Data S2). Likewise, top abundant bacterial genera in SCM were *Lactococcus* (39.20%), *Chryseobacterium* (23.55%), *Ralstonia* (6.65%), *Serratia* (4.43%), *Escherichia* (4.98%), *Riemerella* (2.58%), *Pseudomonas* (1.67%), *Citrobacter* (1.40%), *Streptococcus* (1.13%), *Pedobacter* (1.08%) (Fig. 3). Conversely, the H-milk was mostly dominated by two bacterial genera, *Acinetobacter* (73.66%) and *Pseudomonas* (24.01%), with relatively lower abundances of *Psychrobacter* (0.62%), *Streptococcus* (0.25%), *Lactococcus* (0.22%), and *Ralstonia* (0.18%) (Fig. 3, Data S2). Despite having substantially lower (< 0.5) relative abundances, 60, 7, 3

and 27 exclusively associated bacterial genera were found in CM, RCM, SCM and H-milk samples (Data S2). These findings suggest that mastitis-associated microbiomes have higher relative abundances of the detected bacterial genera compared to the H milk microbiome, with the inclusion of new or the same genera of opportunistic bacteria. However, the relative abundances of mastitis causing bacteria (for example: *Escherichia*, *Klebsiella*, *Shigella*, *Salmonella*, *Enterobacter*, *Shewanella*, *Pantoea*, *Citrobacter*, *Aeromonas*, *Cronobacter*, *Chryseobacterium*, *Serratia*, *Riemerella*, and *Pedobacter*) remain more inconsistent according to the different categories of mastitis across CM, RCM and SCM (Data S2).

We further investigated the strain-level differences of microbial communities across these four levels of mastitis through PS analysis which revealed significant variations ($p = 0.002$, Kruskal-Wallis test) in microbiome composition, diversity and relative abundances (Fig. 4, Data S2). The strain-level microbiome shift was also evident within three mastitis conditions, with a significantly ($p = 0.001$, Kruskal-Wallis test) higher number of bacterial strains in the CM-microbiome over the RCM and SCM. Of the detected strains, 62.33%, 13.85% and 17.50% had a sole association with bovine CM, RCM and SCM, respectively (Fig. 2a). However, the presence of few predominating bacterial species in the four microbiomes (particularly in RCM and SCM) suggests that the crucial differences might also be occurring at the strain-level, and most of the species identified in each sample of the corresponding microbiome were represented by a single strain. The

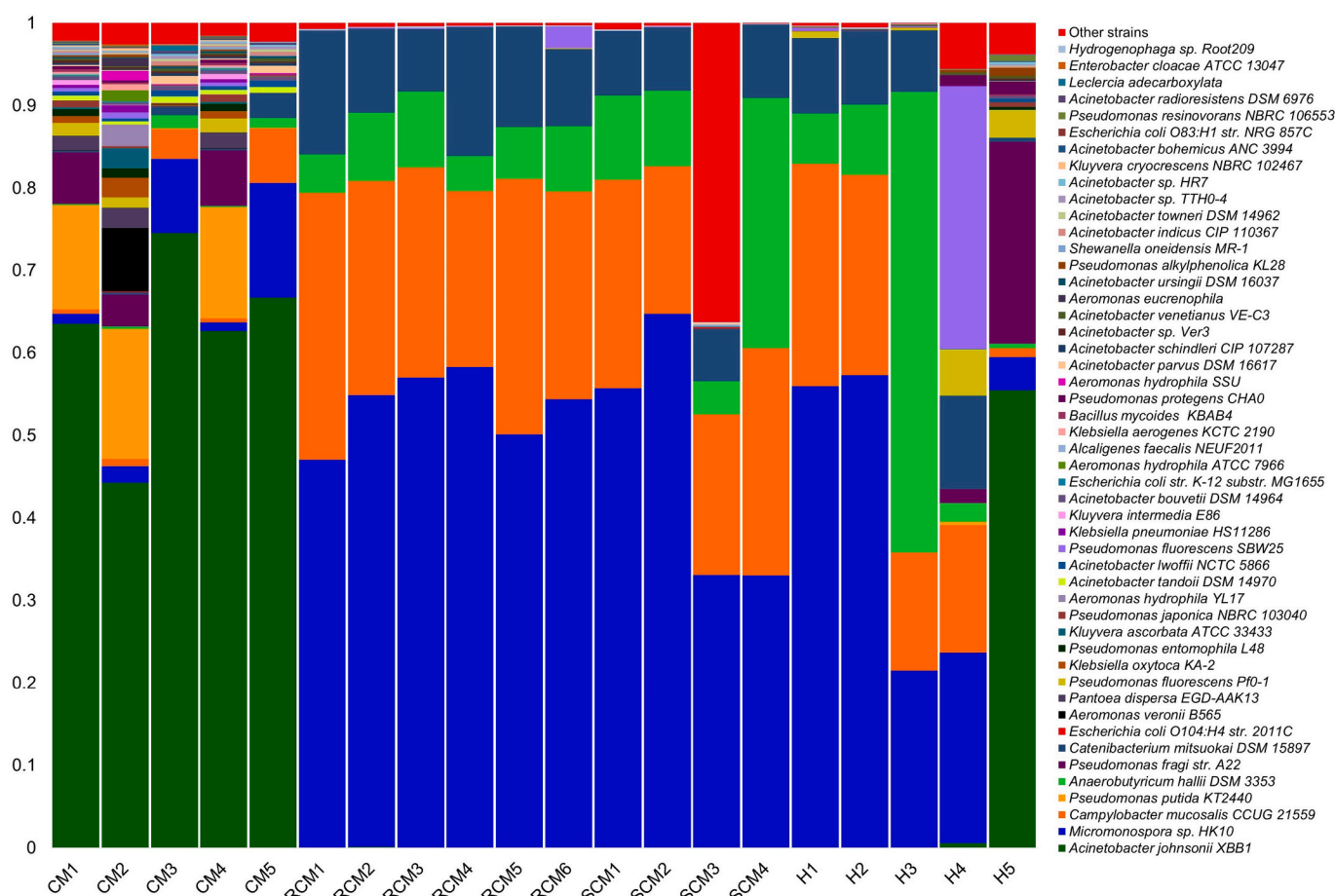


Fig. 4. The strain level taxonomic profile bacteria in bovine mastitis and healthy milk. Stacked bar plots showing the relative abundance and distribution of the 50 most abundant strains, with ranks ordered from bottom to top by their increasing proportion among the CM, RCM, SCM and H milk samples. Only the 49 most abundant strains are shown in the legend, with the remaining strains grouped as 'Other strains'. Each stacked bar plot represents the abundance of bacteria in each sample of the corresponding category. The distribution and relative abundance of the bacterial strains in four metagenomes are also available in Data S1.

most prevalent bacterial strain, *Acinetobacter johnsonii* XBB1 had almost 1.7-fold higher relative abundance in CM (59.92%) milk over H (36.2%) milk but remained much lower ($< 0.5\%$) in the other two microbiomes (RCM, SCM) (Fig. 4, Data S2). Moreover, *Pseudomonas putida* KT2440 (11.81%), *Escherichia coli* O104:H4 str. 2011C (0.036%), *Aeromonas veronii* B565 (1.79%), *Pantoea dispersa* EGD-AAK13 (1.74%), *Klebsiella oxytoca* KA-2 (1.09%), *Pseudomonas entomophila* L48 (0.80%), *Kluyvera ascorbata* ATCC 33433 (0.68%), *P. japonica* NBRC 103040 (0.64%), *Aeromonas hydrophila* YL17 (0.61%) and *Acinetobacter tandoii* DSM 14970 (0.56%) were the predominant strains found in CM milk (Fig. 4, Data S2). *Micromonospora* sp. HK10 (54.11%), *Campylobacter mucosalis* CCUG 21559 (26.52%), *Catenibacterium mitsuokai* DSM 15897 (11.15%), and *Anaerobutyricum hallii* DSM 3353 (7.09%) were the predominant strains in RCM. The most abundant strains in SCM were *Micromonospora* sp. HK10 (38.29%), *Campylobacter mucosalis* CCUG 21559 (26.25%), *Anaerobutyricum hallii* DSM 3353 (25.74%), and *Catenibacterium mitsuokai* DSM 15897 (8.62%). In contrast, the H-milk microbiome was mostly dominated by *Acinetobacter johnsonii* XBB1 (36.20%), *Micromonospora* sp. HK10 (15.16%), *Anaerobutyricum hallii* DSM 3353 (11.60%), *Campylobacter mucosalis* CCUG 21559 (7.42%), and different strains of *Pseudomonas* such as *Pseudomonas fragi* str. A22 (16.04%), *Pseudomonas fluorescens* SBW25 (2.43%), *Pseudomonas protegens* CHA0 (1.02%), *Pseudomonas fluorescens* SBW25 (0.70%), *Pseudomonas alkylphenolica* KL28 (0.65%), and *Pseudomonas resinovorans* NBRC 106553 (0.52%). The rest of the bacterial strains identified in these samples had much lower ($< 0.50\%$) relative abundances (Fig. 4, Data S2).

The literature on the strain-level microbiome composition in bovine mastitis through shotgun metagenomics is scarce [6,14], and most of the microbiome studies carried out in bovine mastitis are restricted only up to detection of bacteria at genus level using the amplicon sequencing. Therefore, most bacterial strains identified in this WMS study are yet to be reported (referred to as unreported). Across the three mastitis conditions, 67.19% of the detected bacterial strains were previously unreported in bovine mastitis (of which 70.13% were in CM, 44.62% were in RCM and 43.75% were in SCM), and possibly played an opportunistic role in mammary gland pathogenesis. The top abundant opportunistic strains in CM-milk were *Aeromonas veronii* B565 (19.16%), *Pantoea dispersa* EGD-AAK13 (18.65%), *Klebsiella oxytoca* KA-2 (11.70%), *Kluyvera ascorbata* ATCC 33433 (7.27%), *Aeromonas hydrophila* YL17 (6.54%). In RCM-milk, the most abundant strains were *Nocardia pseudobrasiliensis* (48.06%), *Serratia marcescens* subsp. *marcescens* Db11 (16.22%), *Nocardia mikamii* NBRC 108933 (15.19%), *Corynebacterium bovis* DSM 20582 (4.66%), and *Aeromonas hydrophila* SSU (3.11%). Different strains of the *Chryseobacterium* genus such as *Chryseobacterium* sp. Leaf405 (19.06%), *Chryseobacterium* sp. CF299 (13.68%), *Chryseobacterium haifense* DSM 19056 (13.19), *Chryseobacterium greenlandense* (13.04%), and *Chryseobacterium* sp. YR460 (12.56%), and *Serratia marcescens* subsp. *marcescens* Db11 (8.30%) and *Citrobacter freundii* CFNIH1 (6.65%) were the most abundant opportunistic strains in SCM (Table S2).

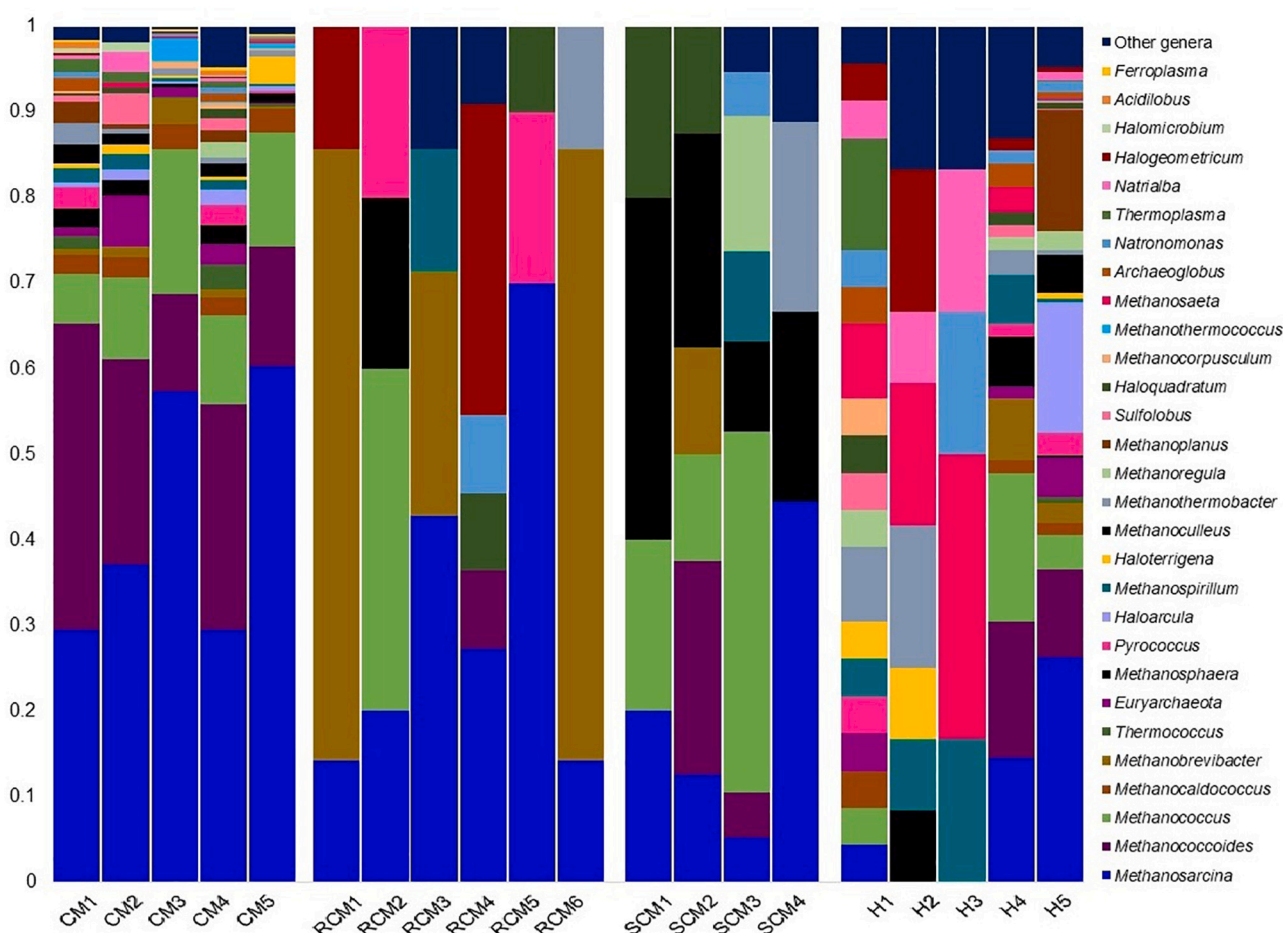


Fig. 5. The taxonomic profile archaea in bovine mastitis and healthy milk. Bar plots showing the distribution and relative abundance of top 30 most abundant archaeal genera, with ranks ordered from bottom to top by their increasing proportion among the CM, RCM, SCM and H milk samples. Only the 29 most abundant genera are shown in the legend, with the remaining strains grouped as 'Other genera'. Each stacked bar plot represents the abundance of archaea in each sample of the corresponding category, and notable differences in the archaeal populations are those where the taxon is abundant in one sample category and effectively undetected in the other sample types. The distribution and relative abundance of the archaeal genera in four metagenomes are also available in Data S1.

2.3. Mastitis-associated shift in archaeal and viral fraction of the microbiome

Another noteworthy feature of the current study is the concomitant detection of archaeal and viral fractions of microbiomes in both mastitis and H-milk samples through MR analysis. Interestingly, the structure and relative abundance of several microbial clades of these two domains were decreased in both RCM (11 genera) and SCM (9 genera) milk and remained enriched in CM-milk (56 genera) and healthy controls (46 genera) (Figs. 4,5, Data S2). Among the identified archaeal components of the microbiome, *Methanosarcina* was the most abundant genus both in CM (41.78%) and RCM (34.04%) milk samples. Besides this genus, *Methanococcoides* (21.25%), *Methanocaldococcus* (2.43%), *Thermococcus* (1.59%), *Haloterrigena* (0.95%), *Sulfolobus* (0.80%), *Methanothermococcus* (0.71%) and *Methanocorpusculum* (0.59%) in CM-milk, *Methanobrevibacter* (25.53%), *Halogeometricum* (10.64%) and *Pyrococcus* (6.38) in RCM-milk, and *Methanococcus* (24.39%), *Methanosphaera* (19.51%), *Methanoregula* (7.31%), *Methanospirillum* and *Methanothermobacter* (4.88% each) in SCM-milk were the predominantly abundant archaeal genera (Fig. 5). On the other hand, *Haloarcula* (12.94%), *Methanoplanus* (11.95%), *Euryarchaeota* (4.12%), *Methanoculleus* (3.84%), *Methanosaeta* (1.28%), *Archaeoglobus* (1.0%), *Natrialba* and *Halogeometricum* (1.13%, each) were the most abundant archaeal genera in H-milk (Fig. 5). Remarkably, the 81.03% and 77.59% of the detected archaeal genera were absent from the RCM and SCM

microbiomes, respectively (Data S2). The relative abundance of *Haloquadratum* and *Natronomonas* remained more than two-fold higher in SCM and RCM samples than CM and H milk samples (Fig. 5). The relative abundance of the rest of the genera remained much lower ($< 0.05\%$) but varied significantly across the four sample types (Fig. 5, Data S2). Moreover, 13 (23.21%) and 2 (4.35%) archaeal genera in CM and H milk samples, respectively, had unique associations while none were found to be unique in RCM and SCM milk microbiomes. Among these unique genera, *Acidilobus*, *Aciduliprofundum*, *Candidatus*, *Pyrobaculum*, *Halobacterium*, *Thermosphaera* etc. were found in CM while only *Ferroglobus* and *Methanopyrus* were identified in H-milk samples (Data S2).

The viral fraction of the microbiome was largely dominated by the members of the orders *Ortervirales* and *Caudovirales* ($> 90.0\%$ of the total abundance), represented mostly by *Retroviridae* (52.84%), *Siphoviridae* (21.87%), *Podoviridae* (12.96%), *Myoviridae* (11.21%); the remaining families had relatively lower abundances ($< 0.5\%$) (Data S2). The CM milk samples had the highest number (40) of viral genera followed by H (37), RCM (24) and SCM (11) milk samples, and of them, only 27.50% and 35.13% viral genera had sole association in CM and H-milk, respectively. Besides, 60.0%, 27.50% and 45.3% genera in RCM, SCM and H-milk, respectively were found to be shared with CM-milk microbiome (Data S2). In this study, *Siphovirus* was the predominantly identified viral genus across the mastitis samples, with a relative abundance of 50.42%, 24.76%, and 20.33% in SCM, RCM, and CM,

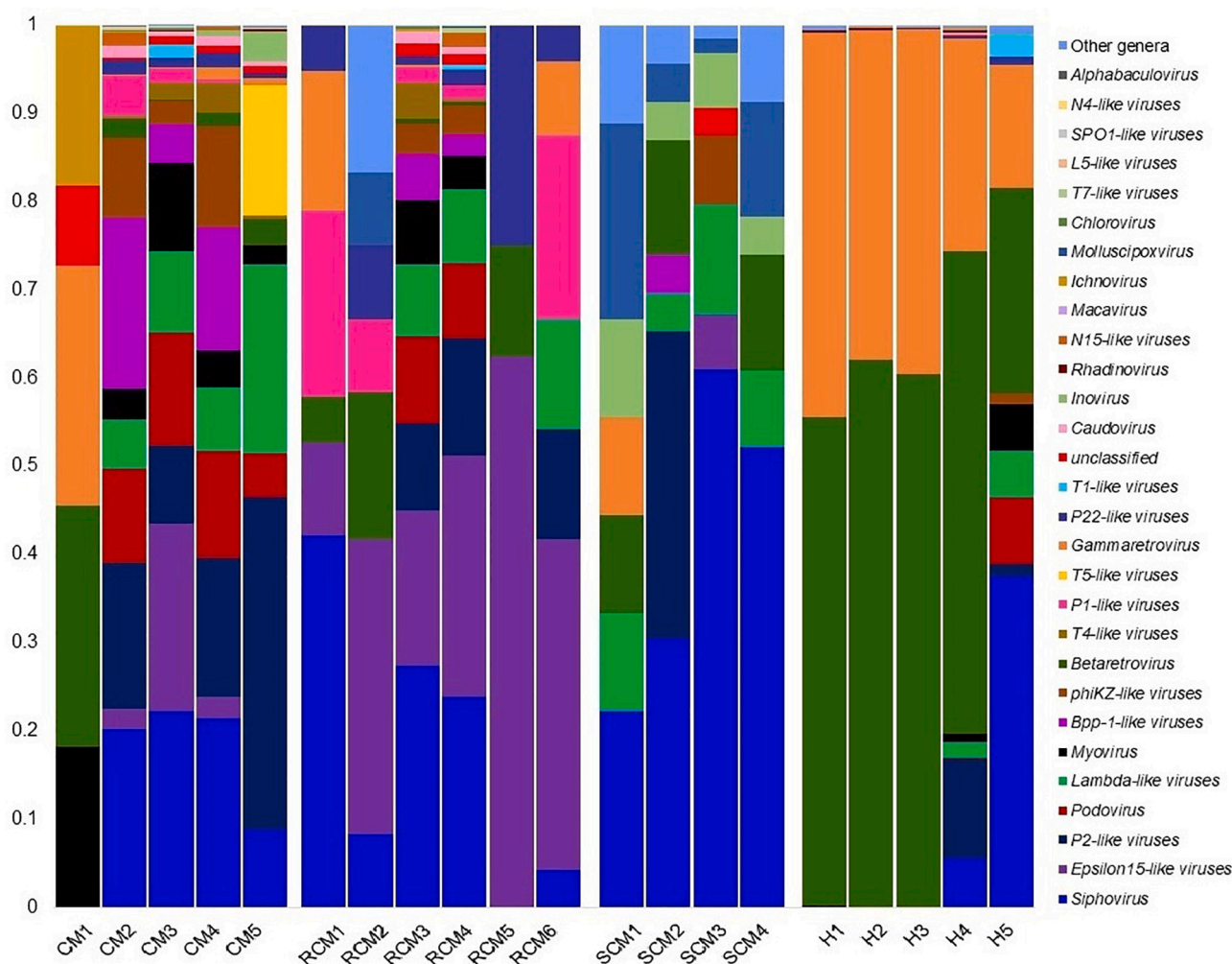


Fig. 6. The taxonomic profile virus in bovine mastitis and healthy milk. Taxonomic distribution of 30 viral genera detected in all of the 20 samples of CM, RCM, SCM and H milk metagenomes. The most abundant viral genera are sorted by descending order of the relative abundance. Each stacked bar plot represents the abundance of viruses in each sample of the corresponding category, and notable differences in the viral populations are those where the taxon is abundant in one sample category and effectively undetected in the other sample types. The distribution and relative abundance of the viral genera in four metagenomes are also available in Data S1.

respectively (Fig. 6, Data S2). In H-milk samples the relative abundance of this genus was only 13.67% (Fig. 6, Data S2). In addition to *Siphovirus*, the most abundant viral genera in CM were *P2-like viruses* (13.67%), *Podovirus* (11.57%), *Myovirus* (7.78%) *Bpp-1-like viruses* (6.60%), *phiKZ-like viruses* (4.06%) and *T1-like viruses* (0.99%). Similarly, *Epsilon15-like viruses* (23.78%), *P1-like viruses* (2.53%), *P22-like viruses* (1.85%), *T4-like viruses* (1.75%), *Caudovirus* (0.97%) and *N15-like viruses* (0.88%) in RCM-milk, and *Lambda-like viruses* (10.08%), *Inovirus* (5.88%), *Molluscipoxvirus* (5.88%) and *phiKZ-like viruses* (4.20%) in SCM-milk were the predominant viral genera (Fig. 6). The H-milk, however, had a relatively higher abundance of *Betaretrovirus* (46.42%) and *Gammaretrovirus* (30.07%) genera. The rest of the viral genera detected in both mastitis and H-milk had much lower (< 0.5%) relative abundances but their abundances always remained higher in CM (Fig. 6, Data S2). Despite having lower relative abundances (< 0.5%), *Epsilonpapillomavirus*, *Gammapapillomavirus*, *Lymphocryptovirus*, *Mastadenovirus*, *Microvirus* etc. in CM-milk, and *AHJD-like viruses*, *Alpharetrovirus*, *Avipoxvirus* c2-like viruses, *Cervidpoxvirus*, *Epsilonretrovirus*, *Lentivirus*, *Parapoxvirus*, *Retrovirus* in H-milk were found as the unique viral genera (Data S2).

2.4. Virulence factor gene families in mastitis and healthy milk

We next sought to identify VFGs among the microbiomes of the four sample categories through mapping the WMS reads against the virulence factor database (VFDB) [33]. The VFDB annotations revealed a clear and distinct enrichment of the predicted VFGs in association with bovine mastitis. We detected 494 VFGs which showed significant ($p = 0.001$, Kruskal-Wallis test) differences both in composition and relative abundances among these four sample categories (Data S3). The CM-associated microbiome had the highest number of VFGs (333), followed by RCM (304), SCM (183) and H (50) milk samples, and only 3.24% of the detected VFGs were shared among the microbiomes of the four mastitis types (Fig. 7a, b). In this study, 77.4%, 10.3% and 13.3% of the detected VFGs in RCM, SCM and H-milk samples, respectively, had a shared association with CM samples (Fig. 7a, Data S3). The virulence properties (number and relative abundances of the VFGs) of the microbiomes (bacterial fraction of microbiomes) were strongly correlated ($p = 0.001$, Kruskal-Wallis test) to their relative abundances of bacteria (at the genus level) in the corresponding CM, RCM, SCM, and H-milk samples (Table S3). By comparing the relative abundance of the detected VFGs among the microbiomes of four samples categories, we found that immune evasion associated capsular genes (*ABZJ_00085* and *ABZJ_00086*) of *Acinetobacter* remained higher in H-milk (10.05%

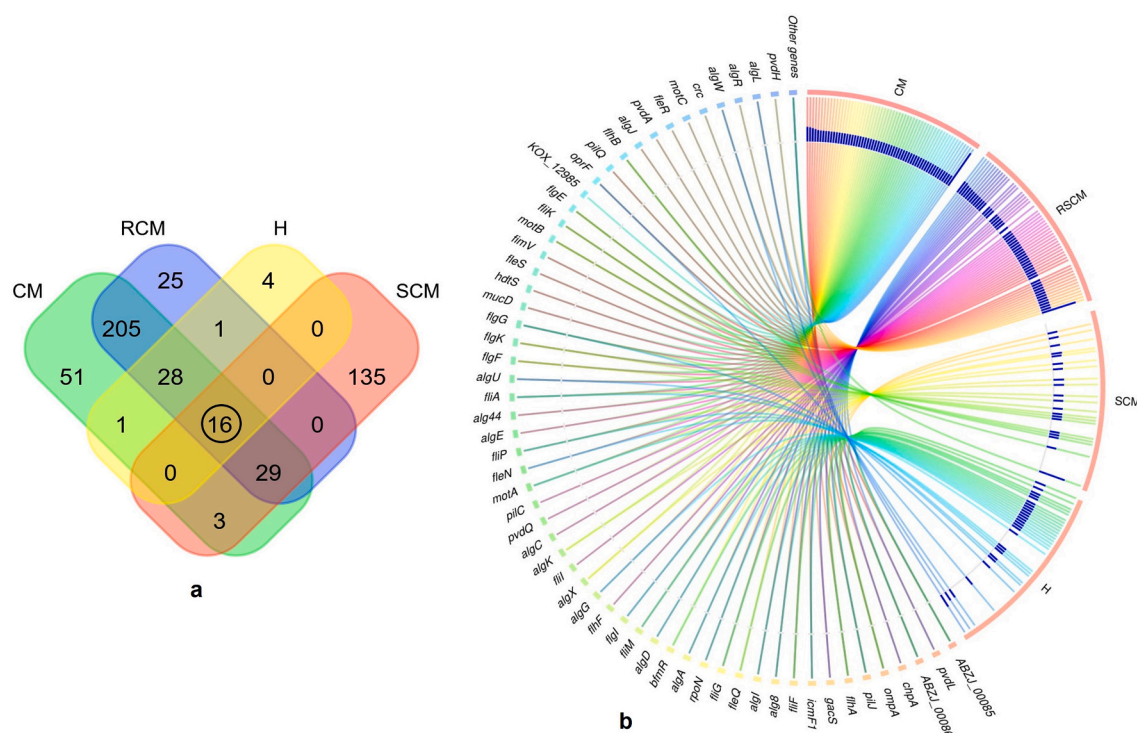


Fig. 7. Virulence factors associated genes (VFGs) detected in bovine mastitis and healthy milk metagenomes. Metagenome sequencing data was used to search for open reading frames (ORFs) compared against the VFDB database to identify the VFGs with over 95% sequence identity. (a) Venn diagram showing unique and shared VFGs detected in CM, RCM, SCM and H milk metagenomes. VFGs sharing between the conditions are indicated by black circle. (b) The circular plot illustrates the distribution of top abundant 50 VFGs found across the four metagenomes. VFGs in the respective metagenome group are represented by different colored ribbons, and the inner blue bars indicate their respective relative abundances. The CM associated microbiomes had the highest number of VFGs followed by RCM, SCM and H metagenomes. The distribution and relative abundance of the VFGs in four metagenomes are also available in Data S2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and 6.87%, respectively) followed by CM (5.58% and 5.26%, respectively), and RCM (4.07% and 3.55%, respectively), corroborating the higher taxonomic abundance of this genus in H-milk. Surprisingly, these two genes were absent in SCM-milk samples (Fig. 7b). Importantly, the CM and RCM-microbiomes had sole association of several VFGs including *pvdL*; crucial nutrient (iron) uptake and/or regulation (5.11% and 5.28%, respectively), *chpA*; twitching motility (3.17% and 2.90%, respectively), *motC*; chemotaxis (0.55% and 0.44%, respectively), *motB/ompA*; bacterial adhesion, invasion, or intracellular survival (2.1% and 2.34%, respectively), *pilJ*; DNA uptake, adhesion to host cells and adherence (1.83% and 1.81%, respectively), *pvdA*; biofilm-formation and quorum-sensing (0.56% and 0.60%, respectively), *gacS*; two-component regulatory system (1.55% and 1.45%, respectively), *icmF1*; intracellular multiplication factor (1.51% in each), *bfmR*; master regulator of biofilm initiation (1.57% and 1.24%, respectively), *pvdQ*; quorum-sensing/antibiotics susceptibility altering (0.88% and 0.96%, respectively), *pilC*; cellular adherence (0.88% and 0.91%, respectively), *mucD*; heat shock or stress-related (0.65% and 0.55%, respectively), *hdtS*; bacterial communication (0.65% and 0.55%, respectively), and *crc*; bacterial pathogenesis (0.54% and 0.24%, respectively) (Fig. 7b, Data S3). The SCM-microbiomes possessed a higher number of genes such as *EcSMS35_B0007*; bacteriocin production and cytotoxic activity in *E. coli* (5.28%), *cpsB*; capsular proteins in pathogenic *Klebsiella* species (2.60%), *rfb*; O-antigen synthesis in Gram-negative bacteria (1.77%), *stgC*; fimbrial proteins (2.10%), and *upaG/ehaG*; adhesin proteins (2.09%) in pathogenic *E. coli* (Fig. 7b, Data S3). Conversely, several genes namely *ompA*; bacterial chemotaxis (3.43%), *motA*; bacterial motility (1.63%), *flhM* (3.08%), *flhA* (2.90%), *flgI* (2.18%), *flhG* (2.13%), *flgG* (1.87%), *flhF* (1.86%), *flhA* (1.47%) and *flhP* (1.41%); flagellar assembly, *alg8* (3.93%), *algI* (2.82%), and *algD* (2.76%); biofilm-formation, *algG* (4.28%) and *algU* (1.62%); alginate biosynthesis,

rpoN (4.87%), *flhQ* (3.30%) and *flhN* (1.69%); pili and flagella expression/adherence, and *oprF*; outer membrane proteins (3.33%) had relatively over expression in H-milk microbiomes compared to mastitis-causing microbiomes (Fig. 7b). Although, the rest of the VFGs detected in this study had relatively lower abundances (< 0.5%), they remained predominantly identified in both the CM and RCM-associated bacteriomes (Data S3).

2.5. Antibiotics resistance gene families in mastitis and healthy milk microbiomes

We further investigated the total number and classes of different antibiotic resistance genes (ARGs) present in the microbiomes of these four sample categories using ResFinder [34]. In our current WMS data set, there was broad variation in ARGs diversity and composition (Fig. 8a, Data S3). The categories and relative abundances of the ARGs were significantly correlated ($p = 0.0001$, Kruskal-Wallis test) with the relative abundance of the associated bacteria found in CM, RCM, SCM, and H-milk samples (Data S3). ResFinder identified 58 ARGs belonged to 18 antibiotic classes distributed in 442 bacterial genomes (Data S3). The CM-associated microbiomes harbored the highest number of ARGs (48), followed by RCM (31), SCM (11) and H (6) milk microbes (Fig. 8a). However, the RCM-causing bacteria possessed the highest number (12) of ARG classes followed by CM (11), SCM (9) and H (4) milk bacteria (Data S3). In addition, 36.2%, 20.4% and 12.5% of the ARGs in RCM, SCM and H-milk microbes, respectively had shared association with those of CM-pathogens (Fig. 8b). Of the resistant classes, beta-lactams (11 genes) were detected as the most prevalent antibiotic group in all samples followed by tetracyclines (5 genes), quinolones (5 genes), macrolides (10 genes), and trimethoprim (3 genes) (Fig. 8b, Data S3). The relative abundances of these ARGs varied across the four

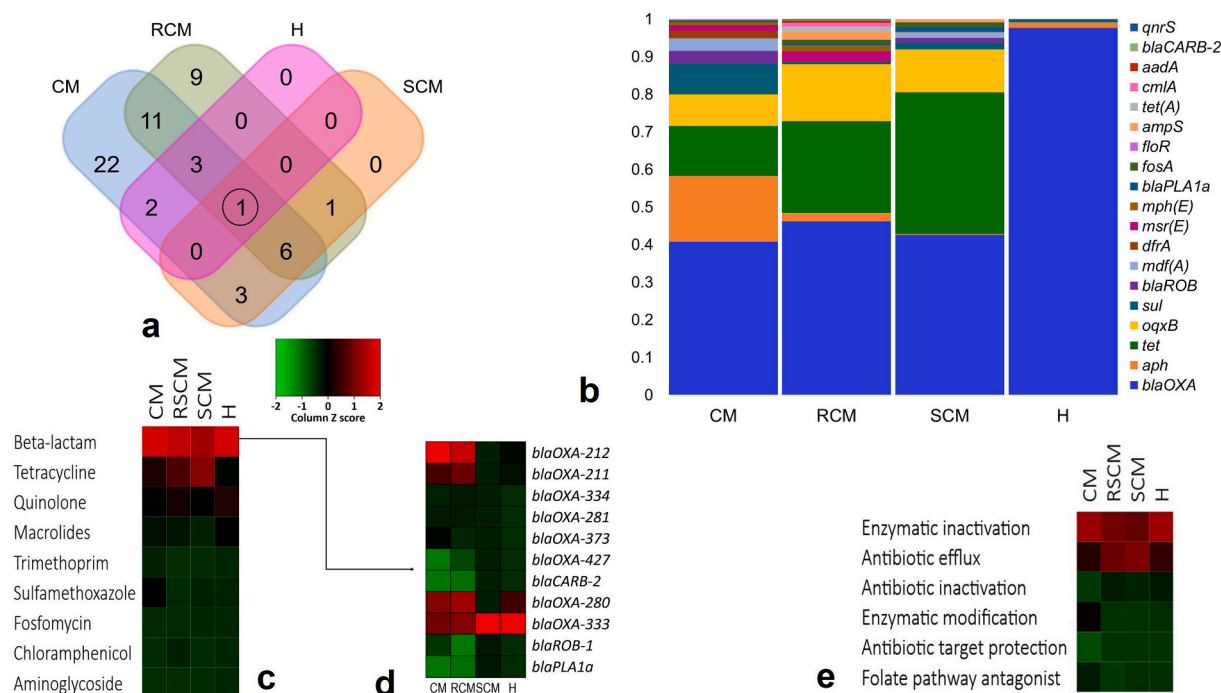


Fig. 8. Antibiotic resistance genes (ARGs) in bovine mastitis and healthy milk metagenomes. Metagenome sequencing data was used to search for open reading frames (ORFs) compared against the ResFinder database to identify ARGs with over 95% sequence identity. (a) Venn diagram showing unique and shared ARGs detected in CM, RCM, SCM and H milk metagenomes. ARGs sharing between the conditions are indicated by black circle. (b) Stacked bar plots showing the relative abundance and distribution of the ARGs across the four metagenomes. (c) Heatmap showing the classes of antibiotics associated with the identified ARGs in bovine mastitis and healthy milk microbiomes. (d) Different beta-lactam genes resistant genes found in four metagenomes. (e) The possible mechanisms of actions/pathways of antimicrobial resistance found in this metagenome sequences. The color code indicates the presence and completeness of each gene, expressed as a value (Z score) between -2 (low abundance), and 2 (high abundance). The red color indicates the highest abundance whilst light green cells accounts for lower abundance of the respective genes in each metagenome. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

samples categories (Fig. 8b, c). The broad-spectrum beta-lactams resistant gene, *bla_{OXA}* was found as the common ARG among the microbiomes of four categories, displaying the highest relative abundance (97.63%) in H-milk samples followed by RCM (46.17%), SCM (42.54%) and CM (40.80%) (Fig. 8b, d). However, other beta-lactams resistant genes such as *bla_{ROB}* in CM (3.73%) and SCM (1.27%), *bla_{PLA1a}* in CM (0.31%), RCM (0.29%) and SCM (1.16%), and *bla_{CARB-2}* in RCM (0.2%) causing pathogens were also identified (Fig. 8d). The tetracyclines (doxycycline and tetracycline) resistant gene, *tet* had the highest relative abundance (37.62%) in SCM-associated bacteria compared to RCM (25.84%) and CM (13.33%)–pathogens and was not detected in the H-milk samples (Fig. 8b). Another tetracycline resistant gene, *tetA* was only found in the RCM milk-microbiome with a relative abundance of 1.39% (Fig. 8b). Aminoglycoside (*aph*) and sulfonamide (*sul*) resistant genes were ubiquitously prevalent among the microbiota of the four sample categories, but their relative abundances remained several-folds (> 5) higher in CM-pathogens compared to RCM, SCM and H milk. Macrolides (erythromycin and streptogramin B) resistant (*msrE*), and fosfomycin resistant (*fosA*) genes had higher relative abundances (2.73% and 1.15%, respectively) in RCM-pathogens compared to other mastitis causing bacteria. The multidrug-resistant gene, *oqx(B)* had the highest relative abundance in RCM (15.11%) while *mdfA* remained predominantly abundant in CM (3.31%) pathogens (Fig. 8b). Interestingly, the *dfrA* (trimethoprim resistant, 2.01%) and *floR* (florfenicol resistant, 0.24%) genes were identified in only the CM-associated microbial genomes (Fig. 8b). Similarly, the RCM-microbiome also had a unique association with four ARGs; the *cmlA* (chloramphenicol resistant), *aadA* (aminoglycoside resistant), *bla_{CARB-2}* (beta-lactam resistant), and *qnrS* (quinolone resistant) genes which were absent among the microbes of other samples (Fig. 8b, Data S3). In general, beta-

lactam resistant genes (*bla_{OXA}* particularly) presented the highest level of expression (Fig. 8c, d) followed by tetracycline (*tet* and *tetA*), quinolone (*oqx(B)*, *mdfA* and *qnrS*), and macrolide (*mphE*, *msrE* and *mdfA*) resistant genes. The rest of the ARGs also varied in their expression levels across the four categories, being more prevalent in the RCM-microbiomes (Fig. 8c). By investigating the possible mechanisms of the detected ARGs, we found that enzymatic inactivation associated resistance had the highest level of expression in the microbiomes followed by antibiotic efflux pumps, antibiotic inactivation, enzymatic modification, antibiotic target protection/alteration, and folate pathway antagonist-attributed resistance mechanisms (Fig. 8e, Data S3).

2.6. Mastitis-associated changes in metabolic functional potentials of the microbiome

The functional annotation of the WMS reads identified 8.55% and 5.98% putative genes with known and unknown protein functions, respectively (Data S2). By examining the correlation between the different gene families of the same KEGG pathway for mastitis and H-milk microbiomes, we found significant differences ($p = 0.002$, Kruskal-Wallis test) in their relative abundances and positive correlation with different conditions of bovine mastitis. We found that genes coding for citrate metabolism (TCA cycle) had around two-fold overexpression in the CM (16.55%) and SCM (12.50%)-associated microbiomes compared to RCM (6.75%) and H (8.79%)-milk. In addition, genes encoding citrate synthase (*gltA*) had higher relative abundance in the RCM-microbiomes (31.25%), followed by SCM (21.57%), CM (15.41%) and H-milk microbes (13.86%) (Fig. 9a, Data S3). The RCM and SCM-milk microbiomes had higher relative abundances of genes associated with oxidative phosphorylation; OP (74.04% and 78.86%, respectively), and

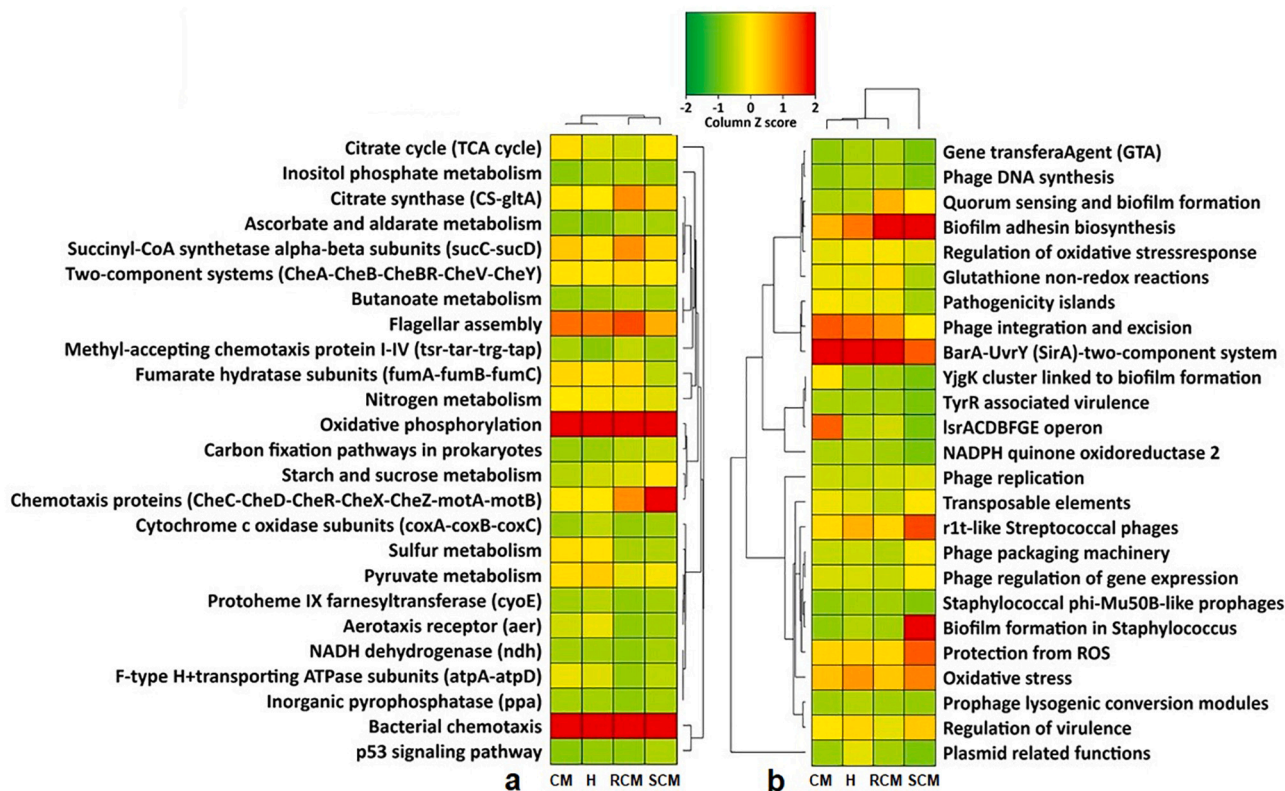


Fig. 9. Projection of the mastitis and healthy milk metagenomes onto KEGG pathways and SEED subsystems. The WMS reveals differences in metabolic functional pathways in CM, RCM, SCM and H milk microbiomes. (a). Heatmap showing the average relative abundance hierarchical clustering of the predicted KEGG Orthologs (KOs) functional pathways of the microbiota across all samples. (b). Comparison of metagenomic profiles of CM, RCM, SCM and H milk microbiotas at different levels of SEED subsystems (up to functions). The selected subsystems showing significant ($p < 0.05$) differences among four sample groups is shown. The color bar at the top represents the relative abundance of putative genes. The color code indicates the presence and completeness of each KEGG and SEED module, expressed as a value (Z score) between -2 (low abundance), and 2 (high abundance). The red color indicates the highest abundant patterns of KOs and SEED subsystems, whilst light green cells accounts for less abundant KOs and SEED modules in that particular sample. More information on metabolic functional potentials (KOs and SEED modules) of the microbiomes in four metagenome groups are available in Data S2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

carbon fixation pathways in prokaryotes (4.81% and 5.71%, respectively) (Fig. 9a). The genes related to nitrogen (13.67%) and sulfur (16.47%) metabolism remained predominantly abundant in CM-microbiomes (Fig. 9a). Furthermore, within OP metabolic pathways, the predicted gene families of NADH dehydrogenase (*ndh*), ATPase subunits (*atpA* and *atpD*), and inorganic pyrophosphatase (*ppa*) had higher relative abundances in CM-microbiomes. The gene or protein families associated with methyl-accepting chemotaxis (*tsr*, *tar*, *trg* and *tap* genes) had higher relative abundances in RCM (5.56%) and CM (4.13%)-causing pathogens. Notably, genes responsible for succinyl-CoA synthetase alpha-beta subunits (*sucC*, *sucD*) (31.25%), and two-component systems (*CheA*, *CheB*, *CheBR*, *CheV*, *CheY*) (18.52%) remained over-expressed in RCM-microbiomes (Fig. 9a). Conversely, the H-milk microbiomes had comparative overexpression in genes coding for pyruvate metabolism (19.31%), cytochrome c oxidase subunits such as *coxA*, *coxB*, and *coxC* (5.47%), and protoheme IX farnesyltransferase, *cyoE* (4.77%) (Fig. 9a, Data S3).

We also sought to gain further insight into the SEED hierarchical protein functions represented by different genes and found 25 statistically different ($p = 0.001$, Kruskal-Wallis test) SEED functions in mastitis (CM, RCM, SCM) and H-milk metagenomes. Overall, the mastitis associated microbiome showed a higher relative abundance of these SEED functions compared to H-milk microbiomes, except for plasmid related functions (highest in H microbes; 6.90%) (Fig. 9b, Data S3). The BarA-UvrY (*SirA*)-two-component regulatory system was found as the most abundant functional pathway in RCM-pathogens (100.00%) followed by CM (85.60%), H (83.66%) and SCM (33.33%)-milk

microbiomes (Fig. 9b, Data S3). The CM-microbiomes were enriched in genes coding for phage integration and excision (36.16%), autoinducer-2 (AI-2) transport and processing (*IsrACDBFGE* operon, 34.74%), pathogenicity islands (15.28%), *YjgK* cluster-linked to biofilm-formation (15.04%), NADPH quinone oxidoreductase-2 (3.34%) and *TyrR* associated virulence (2.93%) compared to the microbes of the other mastitis categories. The RCM-microbiomes however had a higher abundance of SEED functions involved in biofilm adhesion biosynthesis (80.0%), quorum-sensing and biofilm-formation (29.41%), glutathione non-redox reactions (20.83%), regulation of oxidative stress (ROS) response (12.50%), transposable elements (12.06%), phage DNA synthesis (2.38%), and gene transferring agent (1.89%). The SCM-microbiomes particularly showed the elevated expression of genes related to ROS (33.33%), oxidative stress (28.57%), virulence regulation (19.35%), phage related functions such as *r1t*-like *Streptococcal* phages (35.59%), phage regulated gene expression (12.71%), phage packaging machinery (11.86%), phage replication (9.32%) and prophage lysogenic conversion (2.54%) modules. In contrast, most of these SEED modules had relatively lower abundance in H-milk microbiomes (Fig. 9b, Data S3).

3. Discussion

The current study represents the first ever proof-of-concept to decipher dynamics changes in microbiome composition and abundances in CM, RCM, SCM and H-milk samples through the cutting-edge WMS technology. The present findings of higher taxonomic resolution and predicted protein functions, partially consistent with our previous

report [6], and many of the recent 16S rRNA partial gene-based studies [12,13,30]. The microbiome diversity (alpha and beta diversity) measures provided that microbial dysbiosis is closely linked to different stages of mastitis. In this study, we observed increased microbial diversity and species richness in CM, RCM, and SCM-microbiomes relative to healthy-controls. Beta diversity [6,22] also revealed a substantial microbial disparity between healthy-controls and mastitis-milk samples keeping the closest relationship of the CM-microbiomes to RCM followed by SCM, and H-microbiomes. Regardless of higher taxonomic abundances, the bovine mastitis-associated microbiome remained inconsistent and fluctuates more within CM, RCM, and SCM-samples than those of H-milk, agreeing with several recent studies [6,10,11].

In bovine mastitis, predominantly identified pathogens are bacteria [6,7,10–15,17], however, other concomitant microbial players like archaea and viruses [6,11] could also be detected, highlighting the novel insights from microbiome characterizations via metagenomics compared to amplicon sequencing. The bovine milk microbiome was dominated by *Proteobacteria*, *Firmicutes*, *Actinobacteria* and *Bacteroidetes*, and their relative abundance also varied across the samples of four mastitis categories. These results are in line with bovine [6,7,10] and human [22] mastitis microbiome results reported earlier. Previously, *Proteobacteria* and *Firmicutes* were reported as the major phyla in bovine SCM [15] supporting our current findings. During the pathophysiology of mastitis, mammary epithelial cells reduce their capacity to beta-oxidation which alters the composition of indigenous microbiome, reducing the number of anaerobic *Firmicutes* and allowing the proliferation of different genera and/or strains belonging to *Proteobacteria* blooms [7,10,11,36,37].

Metagenomics [6,7,12,13,15] and isolate sequencing [17] of mastitis pathogens provided overwhelming evidence that microbial traits, for instance, pathogenicity, virulence, antibiotic resistance, and metabolic potentials are linked with strain-specific genomic characteristics [23,38–40]. We demonstrated that microbial dysbiosis in different types of mastitis resulted in the depletion of beneficial microbes and enrichment of the opportunistic pathogens. In this study, different strains of *Acinetobacter* were detected as the predominantly abundant CM and RCM-causing pathogen, as also previously reported in human [41], bovine [6], and bubaline [42] mastitis. The CM and RCM samples also had a higher abundance of strains of the *Pseudomonas* which can potentially cause opportunistic infections [6,11]. However, in the SCM samples, the abundance of different strains of *Acinetobacter* and *Pseudomonas* remained substantially lower compared to CM and RCM milk samples. We also revealed that different strains of *Escherichia*, *Aeromonas*, *Klebsiella*, *Shigella*, *Enterobacter*, *Salmonella*, *Pantoea*, *Citrobacter*, *Shewanella*, *Lactococcus*, and *Cronobacter* routinely colonize the mammary tissue or quarters, and manifest different classes of mastitis [6,11,27]. The SCM samples were predominated by different strains of *Lactococcus*, *Chryseobacterium*, *Ralstonia*, *Serratia*, *Riemerella*, *Streptococcus*, and *Pedobacter* corroborating the findings of several earlier reports [10,11,15]. Many of these microbiomes may act as potential opportunists by interfering with metabolism, host defense, and immune development [43–46] to producing different magnitude of udder infections. Our present findings could be further supported by the potential existence of endogenous entero-mammary pathway, and through this axis the gut or rumen microbiomes migrate to the mammary gland to manifests different episodes of mastitis [6,27,40]. The healthy-milk microbiome was also dominated by different strains of environmental, gut, and animal skin originating microbes. Though the pathogenic mechanisms of these commensal microbes are largely unknown, they can cause opportunistic infections in the mammary glands and/or quarters with or without varying degrees of clinical episodes by producing different virulence factors [47] particularly in immunocompromised hosts [6,27,44,45].

The state-of-the-art WMS approach also provided an exciting opportunity for investigating the integrated cross-kingdom interactions of

‘multibiome’– traditionally neglected components of the milk microbiome such as archaea and viruses (< 1.0% genes coding for total microbiome) [48,49]. Unlike bacteria, the diversity and composition of archaea [48] and viruses [49] always remained much lower in both mastitis and healthy-milk [6]. The archaeal fraction of the microbiome is mostly represented by the methanogenic and thermophilic genera of *Euryarchaeota* in CM and H-milk, and halophilic archaeal genera in SCM and RCM-metagenomes. Most of the viruses detected in both mastitis and healthy-milk metagenomes are tailed bacteriophages capable of infecting bacteria, and archaea during physiological challenges that can compromise the defense mechanisms of the udder particularly in lactating cows [6,7]. Even though the role of these accompanying microbes in the pathophysiology of bovine mastitis has not been understood within the frames of a typical host-pathogen interaction, these taxa could cease the opportunity during the pathological changes in the mammary glands created by bacteria [48,49]. They may occupy the microenvironments suitable for anaerobic metabolism, make way for opportunistic pathogens (pathobionts), already present in milk, and/or gain secondary access to the mammary gland. This condition might happen when the host is in an unnatural condition or is immune compromised due to a severe negative energy balance at the onset, in and around lactation, and other environmental stress e.g., heat stress [22,23], and follow the similar virulence mechanisms of bacterial pathogens producing more severe and prolonged recurrent mastitis (RCM) [7,48,49]. Furthermore, a higher abundance of *Haloquadratum* and *Natronomonas* in SCM and RCM-milk may possess some accessory capsular switching genes for their adaptation to environmental fluctuations like bacterial pathogens to causing different magnitudes of disease [50].

It is important to note that the mastitis microbiomes harbored a higher number of VFGs than the H-milk microbiomes. The relative abundance these VFGs were correlated with relative abundance of the associated microbes (bacteria). The mastitis (CM, RCM and SCM) associated microbiomes harbored relatively higher numbers of VFGs pyoverdine or antibiotics susceptibility altering, twitching motility and chemotaxis, adherence to mammary epithelial cells and invasion, two-component regulatory system, flagellar assembly, intracellular survival and multiplication, bacteriocins production and cytotoxic activity, energy transduction, biofilm-formation (BF), and quorum-sensing, and siderophores linked to pathogenesis [1,8,27,51–54]. Conversely, virulence genes coding for innate-immune responses or immune evasion intracellular survival in competition, pili and flagella expression, bacterial chemotaxis, and alginate biosynthesis were predominantly abundant in H-milk microbiomes which might play important roles in protecting bacteria from the host-defenses [8,51–54]. The predominantly abundant *ABZJ* genes found in different strains of *Acinetobacter* with special reference to H-milk metagenome are associated with activation of innate immune cells such as macrophage and neutrophil through heme utilization [52, [55]. The underlying mechanisms for microbial colonization in mammary tissue modulated by these VFGs are not well established. However, VFGs can enable microbes to overcome host defense [8], immune-mediated colonization of the mammary tissues by suppressing the regrowth of resident commensal microbiomes [7,27,56], and subsequent inclusion of opportunistic pathogens with the progression, and recurrence of the disease [7,27,56,57].

We found various homologs of ARGs belonging to different protein families among the microbiome of the four mastitis categories, and the composition and abundances of the ARGs remained significantly correlated with the abundances of the related bacteria. Furthermore, the ARGs also varied greatly in different conditions of mastitis (CM, RCM, SCM) supporting the dynamic dysbiosis of microbiomes, their genetic diversity, and selective pressure for the maintenance of ARGs [9,57]. The β -Lactamases, major resistance determinant for β -lactam antibiotics in Gram-negative bacteria [58], remained with extended-spectrum in the mastitis microbiome. However, β -lactamase resistant gene of OXA (oxacillinase) group, *bla*_{OXA} remained predominantly abundant in H-

milk microbes. This ARG which might be involved in hydrolytic activity against multiple antibiotics [9], and sometimes augmented in mastitis pathogens by additional resistance mechanisms of impermeability or efflux system [59]. The tetracyclines resistant genes were found only in CM, RCM and SCM-pathogens, with comparatively higher abundances in SCM samples, and corroborated with several earlier reports [58,59]. In addition, aminoglycosides, trimethoprim, florfenicol, and sulfonamide-resistant genes were predominantly abundant in CM samples, which could be due to drug-specific acquired resistance mechanisms [58]. Higher number and classes of ARGs including macrolides, fosfomycin, and quinolones resistance genes in the RCM samples might be resulted from their infrequent use in CM, and subsequent treatment failure. The possible mechanisms for the detected ARGs include enzymatic inactivation, prevention of antibiotic-bindings to the targets, catalytic activity, folate pathway antagonist-attributed resistances, and efflux pump and/or system conferring resistance to a wide range of antibiotics [9,58,60]. These ARGs can easily spread by their harboring bacteria throughout different dairy environments because of wind and runoff waters, climate change, human activities, and contact with wild animals, in particular, migratory ones [9,58].

The present study found several important predicted metabolic functions that altered in their abundances among CM, RCM, SCM and healthy-milk microbiomes as also reported previously in lactating cows [12], women [22] and mice [46]. Our analysis revealed that similar metabolic features identified in the same KEGG pathway or SEED subsystem varied in different categories of mastitis (CM, RCM and SCM) suggesting their possible association in the early colonization and disease progression [7,18,20]. Recent studies have shown that bacterial metabolites are involved in modulating immune functions of their host [39,52]. The mastitis-associated microbiomes harbored a higher number of genes for nutrient transport [61] and carbohydrate metabolism through TCA-cycle [62]. Moreover, increased benzoate degradation by different strains of *Acinetobacter* and *Klebsiella* through TCA-cycle might promote bacterial growth and virulence factors expression during mammary gland pathogenesis [63]. The methyl-accepting chemotaxis proteins (MCP) or genes (*tar*, *tsr* and *tap*) act through transmembrane receptors that mediate chemotactic response and cell-to-cell communication in certain bacteria, such as *Salmonella* and *E. coli* [59,64]. In our present study, the methyl-accepting transmembrane chemotaxis genes or MCPs were upregulated in RCM and CM-associated pathogens suggesting their role in cell-to-cell communication, early phase attachment to or entry into the udder tissues, and virulence regulation [63]. The aerotaxis receptor gene was over-expressed in H milk microbiomes, which is thought to sense internal redox states and mediate bacterial movement towards optimal oxygen conditions or mediate maximal growth of bacteria [27,63]. Sometimes CM may lead to gangrenous mastitis where cell apoptosis is a common phenomenon. Inhibition of p53 may allow bacteria to subvert the mammary cell cycle control and apoptosis mechanisms. The activation of the tumor repressor protein, p53, appears to play a central role in modulating the expression of various mammary epithelial cells involution responsive genes [60,65]. Thus, increasingly identified p53 signaling in healthy-milk microbiomes might be a response to a stress signal with activation in a specific manner by post-translational modifications which leads to either cell cycle arrest or mammary gland apoptosis [66]. The global regulating unit BarA-UvrY (*SirA*)-two-component system is well conserved in many bacterial species (*γ-Proteobacteria* in particular), and regulate the production of extracellular factors (exoenzymes or toxins), quorum-sensing, motility, and diverse metabolic functions [52,67]. In many mastitis-causing microbes, BarA/UvrY is important for regulation and coordination of pathogenicity and group behaviors [52,67]. The relative overrepresentation of filamentous phage-related genes in mastitis-pathogens may suggest that bacteriophages participate in the horizontal gene transfer among the members of bovine milk microbiome and eventually to mammary gland pathogens [68]. The altered metabolic pathways such as transport and

processing, pathogenicity islands, and biofilm formation (BF) in mastitis-pathogens are associated with a diverse array of virulence mechanisms for mammary gland pathogenesis [27]. Bacterial BF is a strain-specific or genetically-linked trait that can trigger mammary gland pathogenesis by promoting the phagocyte release, proliferation of reactive oxygen and nitrogen species, and transfer of antibiotic resistance [6,9]. Moreover, RCM-microbiomes had overexpression of genes encoding biofilms adhesins [27,51], glutathione non-redox reactions [69], regulation of oxidative stress [70], and transposable elements [71] which play an important role in many chronic-recurrent bacterial infections like bovine RCM.

Bovine milk is a potential reservoir of diverse groups of microbial communities which could be influenced by different factors such as: breed, parity, milk yield, lactation stage, farm management, and feeding [7,11,27]. However, it is very difficult to compare the microbiomes, associated virulence factors, antibiotic resistance genes, and metabolic functions considering all of these predisposing factors in a single study particularly in different geographic locations. These limitations hamper the ability to fully explore the cross-talk among the microbes present in the microbiome that leads to the progression of disease and interaction between the pathogens and the host.

In conclusion, the WMS approach of the present study provided a comprehensive assessment of the dynamic changes in microbiome composition along with their VFGs, ARGs, and metabolic potentials in different pathological categories of bovine mastitis (CM, RCM, SCM), and healthy milk samples. Our results suggest that co-occurrence of VFGs and ARGs possibly play determining roles in the adaptation of different microbes to mastitis compared to healthy milk. Several predicted functional pathways differed between mastitis and healthy controls, perhaps reflecting metabolic changes associated with the progression of mastitis pathogenesis. These findings point towards the development microbiome-based diagnostics and therapeutics for this economically burdened disease. However, future studies should delve further into characterizing microbiome diversity with larger sample sizes, and the inclusion of gut/rumen microbiome sampling in addition to the milk samples for direct testing of microbial transfer across this axis to confirm the shift of microbiome and associated functions.

4. Methods

4.1. Screening for mastitis, and sampling

Details of the study population and collected samples are presented in Table S1. The California Mastitis Test (CMT®, Original Schalm reagent, ThechniVet, USA), a screening test for SCM [19] was used to diagnose different conditions of mastitis (CM, RCM, and SCM) [21]. The CMT scores were categorized in to five grades based on gel formation in the reaction mixture viz. 0 (negative), T (trace, possible infections), 1 (weak positive), 2 (distinct positive), and + 3 (strong positive). The cows having ≥ 2 CMT score along with gross visible signs of mastitis (fibrosis, inflammatory swellings, pain, visible injury or lesion, atrophy of the tissue and teat blindness, and presence of clots, flakes, blood and changes in consistency in milk) were grouped as CM [16]. The CM cows those received antibiotic treatments for 5–7 days, apparently resolved clinical signs and symptoms, and a reinfection of quarter(s) or udder 10–15 days after antimicrobial cure (with or without visible signs of CM, and ≥ 2 CMT score) were defined as RCM. The cows having a CMT score of T (trace) or 1 (weak positive) with no visible physical signs and symptoms, and no changes in milk were considered as positive for SCM [19]. The cows having no history of mastitis either in current or previous lactation, and have had a sound health with good milking records were grouped as H.

A total of 20 milk samples (CM = 5, RCM = 6, SCM = 4, and healthy (H) = 5) from 20 lactating cows at early stages of lactation (within 10–40 days of calving) having parity of 1 to 5 (parity 1 = 2, parity 2 = 8, parity 3 = 6, parity 4 = 2, and parity 5 = 2) were

collected (Table S1). These cows belonged to different dairy farms of the five districts in Bangladesh: Gazipur = 8 (24.19° N, 90.47° E), Dhaka = 3 (23.81° N, 90.41° E), Manikgonj = 3 (23.86° N, 90.00° E), Chattogram = 4 (22.20° N, 91.98° E), Sirajgonj = 2 (24.31° N, 89.57° E) (Table S1). The samples were collected during the summer months of Bangladesh (March to June 2018) having almost same weather condition in the study areas. The sampling patterns followed simultaneous collection both mastitis (one or more), and H (single) milk samples from the same farm. Approximately 15–20 ml of milk from all of the infected quarters of the mastitic cows, and ≥ 2 quarters of the healthy (H) cows (i.e. pulled sample) was collected in a sterile falcon tube during morning milking (8.0–10.0 am), with an emphasis on pre-sampling discard of first few streams, disinfection of teat-ends and hygiene during sampling [13,21]. CM cows which did not recover fully in spite of receiving different conventional antibiotic treatments, with subsequent persistence of CM episodes, are defined as the cows with RMC [20]. The protocol for milk sample collection was approved by the Animal Experimentation Ethical Review Committee (AEERC) of the University of Dhaka under reference number 79/Biol.Sc. The milk samples were then transported to the laboratory and stored at -20°C until DNA extraction.

The microbiome of bovine milk is known to be influenced by different factors such as: breed, farm management, feeding, milk yield, lactation stage, and parity. However, it is very difficult to compare microbiome compositional differences considering all of these influential factors such a different sample from different breeds, different farm (with different managements), first calf and pluriparous cows in a single study, this could be one of the limitations of this study.

4.2. DNA extraction, shotgun metagenomic sequencing, and sequence reads preprocessing

The total genomic DNA (gDNA) content of each milk sample was extracted by an automated DNA extraction platform (Promega, UK) using previously published protocols [6] along with the manufacturer's instructions. The quantity and purity of the extracted DNA was measured with NanoDrop (ThermoFisher, USA) by measuring 260/280 absorbance ratios. Shotgun whole metagenome sequencing (WMS) libraries were prepared with Nextera XT DNA Library Preparation Kit [72] and paired-end (2×150 bp) sequencing was performed on a NextSeq 500 machine (Illumina Inc., USA) at the George Washington University Genomics Core facility. The current metagenomic DNA generated 416.64 million reads with an average of 20.83 million (maximum = 39.75 million, minimum = 2.72 million) reads per sample (Data S2). The resulting FASTQ files were concatenated and filtered through BBDuk [6] (with options $k = 21$, $mink = 6$, $ktrim = r$, $ftm = 5$, $qtrim = rl$, $trimq = 20$, $minlen = 30$, $overwrite = true$) to remove Illumina adapters, known Illumina artifacts and phiX. Any sequence below these thresholds or reads containing more than one 'N' were discarded. On average, 19.28 million reads per sample (maximum = 37.36 million, minimum = 2.50 million) passed the quality control step (Data S2).

4.3. Taxonomic mapping, classification, diversity, and community analysis

Shotgun whole metagenome sequencing (WMS) data were analyzed using both mapping-based and assembly-based hybrid methods of PathoScope 2.0 (PS) [31,73] and MG-RAST (MR) [32], respectively. In PS analysis, a 'target' genome library was constructed containing all bacterial sequences from the NCBI Database (https://en.wikipedia.org/wiki/National_Center_for_Biotechnology_Information) using the PathoLib module [31,72]. The reads were then aligned against the target libraries using the Bowtie 2 algorithm [74] and filtered to remove the reads aligned with the cattle genome (bosTau8) and human genome (hg38) as implemented in PathoMap (very sensitive local -k 100 -score-min L,20,1.0). Finally, the PathoID [75] module was applied to obtain

accurate read counts for downstream analysis. In these samples, 16.88 million reads (4.05% of total reads) were mapped to the target reference genome libraries after filtering the cow and human genome (Data S2). The raw sequences were simultaneously uploaded to the MR server (release 4.0) with properly embedded metadata and were subjected to quality filtering with dereplication, and host DNA removal (filtering against the set reference genome of both cattle and human), and finally screening for taxonomic and functional assignment. Alpha diversity (diversity within samples) was estimated using the observed species, Chao1, ACE, Shannon, Simpson, and Fisher diversity indices [76] for both PS and MR read assignments and counts. To visualize differences in bacterial diversity, principal coordinate analysis (PCoA) based on the Bray-Curtis distance method [77] for MR data (genus level), and non-metric multidimensional scaling (NMDS) measured by weighted-UniFrac distance on PS data (at strain level) through Phyloseq R package (version 3.5.1) [78] were performed. Taxonomic abundance was determined by applying the "Best Hit Classification" option using the NCBI database as a reference with the following settings: maximum e-value of 1×10^{-30} ; minimum identity of 95% for bacteria, 60% for archaea and viruses, and a minimum alignment length of 20 as the set parameters. The phylogenetic origin of the metagenomic sequences was projected against the NCBI taxonomic tree and determined by the lowest common ancestor (LCA) with the same cutoff mentioned above.

4.4. Virulence factors-associated genes (VFGs) analysis

The virulence factor database (VFDB) and DNA sequences with full datasets for virulence factors of pathogenic bacteria [33] were used to identify virulence factors associated genes (VFGs) among the microbiomes of the four sample categories. Every protein included in each sample category was used as a query to search for similarities to VFG protein-coding traits. Furthermore, we computed the representative counts for the different gene families that coded for VFGs traits from all the generated alignments between our metagenomic query cohort and the preceding databases. Hence, we aimed at retrieving the best hit (best-scored alignment) that enabled us to assign a VFG function to each of the metagenomic proteins. The VFGs which fulfilled the following similarity criteria (cut off): e-value $< 1e^{-5}$, percent identity $\geq 80\%$, alignment length/subject length ≥ 0.8 , and alignment length/query length ≥ 0.8 were included in the study [30]. Thus, the amounts of different classes (gene families) that are present in each metagenome represent its diversity in terms of VFGs traits. We used OmicCircos (version 3.9) [79], an R package based on a Python script for circular visualization of both diversity and composition of virulence factors associated genes (VFGs) across the four sample categories under study.

4.5. Antibiotic resistance genes (ARGs) analysis

Similarly, to identify the antibiotics resistance genes (ARGs) among the microbiomes of all sample categories, we used the ResFinder 2.0 database [34]. The ResFinder database was integrated within AMR++ pipeline [34] to identify the respective genes and/or protein families. Every protein included in each of the addressed metagenomic samples was used as a query to search for similarities to ARGs protein-coding traits. Hence, we aimed at retrieving the best hit (best-scored alignment) that enabled us to assign an ARG function to each of the aforementioned metagenomic proteins. The ARGs which fulfilled the following similarity criteria (cut off): e-value $< 1e^{-5}$, percent identity $\geq 80\%$, alignment length/subject length ≥ 0.8 , and alignment length/query length ≥ 0.8 were included in the study [31]. Thus, the amounts of different classes (gene families) that are present in each metagenome represent the diversity in terms of ARGs traits. The z-score for ARGs was calculated using the mean and standard deviation of the abundances of all individual gene in the four sample categories.

4.6. Functional profiling of the microbiomes

To perform taxonomic functional classification, the reads were mapped onto the Kyoto Encyclopedia of Genes and Genomes (KEGG) database [35], and SEED subsystem identifiers, respectively, on the MR (version 4) metagenome analysis server [32] using the partially modified set parameters (*e*-value cutoff: 1×10^{-30} , min. % identity cutoff: 60%, and min. Alignment length cutoff: 20).

4.7. Statistical analysis

The non-parametric test Kruskal-Wallis rank sum test was used to evaluate differences in the relative percent abundance of taxa in mastitis (CM, RCM, SCM) and H groups for PS data. Comparative taxonomic and functional profiling was performed with the reference prokaryotic metagenomes available in MG-RAST database for statistical analyses. The gene counts were normalized by dividing the number of gene hits to individual taxa/function by total number of gene hits in each metagenome dataset to remove bias due to differences in sequencing efforts. To identify differentially abundant SEED or KEGG functions (at different KEGG and SEED subsystems levels), virulence factors (VFGs), and antimicrobial resistance (ARGs) in four metagenomes, statistical tests were applied with non-parametric test Kruskal-Wallis rank sum test using IBM SPSS (SPSS, Version 23.0, IBM Corp., NY USA).

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ygeno.2020.09.039>.

Data availability

The sequence data reported in this paper are available in the NCBI database (BioProject PRJNA529353) and European Nucleotide Archive (BioProject, PRJEB31210).

Ethics statement

The protocol for milk sample collection from lactating dairy cows was approved by the Animal Experimentation Ethical Review Committee (AEERC), Faculty of Biological Sciences, University of Dhaka (under reference number: 79/Biol.Scs).

Author contributions

MNH, MS, and MAH conceived and designed the study. MNH surveyed and collected field samples. MNH and AI carried out laboratory works including DNA extractions and sequencing. MNH, AI, MSR, MRI and KAC conceived, designed, and executed the bioinformatics analysis. MNH interpreted the results, visualized the figures, and drafted the manuscript. MAH and KAC contributed chemicals and reagents. AA, MS, AMAMZS, KAC and MAH contributed intellectually to the interpretation and presentation of the results. Finally, all authors have approved the manuscript for submission.

Additional Information

Supplementary information supporting the findings of the study are available in this article as Data S1, S2 and S3 files, and Table S1–S3.

Declaration of Competing Interest

The authors declare no competing interests.

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